

Comparison Table

	Intel Core i9-11980HK	AMD Ryzen 9 5900HX	AMD Ryzen 9 5980HS	Intel i7 11800H	Intel i7 11850H	AMD Ryzen 7 5800x	Qualcomm Snapdragon 8cx Gen2 5G Compute Platform	Apple M1	Mediatek MT8192	Intel Xeon W-10885M
Launch year	2021	2021	2021	2021	2021	2020	2020	2020	2021	2020
No. of cores	8	8	8	8	8	8	8	8	8	8
No. of threads	16	16	16	16	16	16	-	8	8	16
Base clock frequency	2600 MHz	3.3GHz	3.0GHz	1900 MHz	2100 MHz	3.8GHz	N/A	2.1 GHz	2.0 GHz	2.4 GHz
Max boost clock frequency	5.00 GHz	Up to 4.6GHz	Up to 4.8GHz	4.60 GHz	4.80 GHz	Up to 4.7GHz	N/A	3.2 GHz	2.2 GHz	5.30 GHz
Cache	24MB	L2 Cache: 4MB, L3 Cache: 16MB	L2 Cache: 4MB, L3 Cache: 16MB	24MB	24MB	L2 Cache: 4MB, L3 Cache: 32MB	N/A	L1 Cache: 320KB per core, L2 Cache: 12MB	1MB	16MB
TDP	Configurable (Down: 45W, Up: 65W)	Configurable (35-54W)	35W Default	Configurable (Down: 35W, Up: 45W)	Configurable (Down: 35W, Up: 45W)	105W Default	N/A	14W	10W	Configurable (Down: 35W, Up: 45W)
Memory	128 GB	DDR4 - Up to 3200MHz	DDR4 - Up to 3200MHz	128 GB	128 GB	DDR4 - Up to 3200MHz	LPDDR4x up to 2133MHz	16GB LPD-DR4X-4266	4GB LPD-DR4X-2133	128 GB DDR4-2933 up to 2133MHz
Maximum number of memory channels	2	N/A	N/A	2	2	N/A	8	2	2	2
Graphics frequency	1.45 GHz	2100 MHz	2100 MHz	1.45 GHz	1.45 GHz	2100 MHz	N/A	N/A	850 MHz	1.25 GHz
Graphics architecture	Intel UHD Graphics	AMD Zen3 Core	AMD Zen3 Core	Intel UHD Graphics	Intel UHD Graphics	AMD Zen3 Core	Qualcomm Adreno GPU	Apple M1 GPU	ARM Mali-G57 MC5	Intel UHD Graphics
No. of graphics core	N/A	8	8	N/A	N/A	8	N/A	N/A	5	N/A
Socket/Package	FCBGA1787	FP6	FP6	FCBGA1787	FCBGA1787	AM4	N/A	Apple M-Socket	SoC	FCBGA1440
CMOS technology	10 nm SuperFin	TSMC 7nm FinFET	TSMC 7nm FinFET	10 nm SuperFin	10 nm SuperFin	TSMC 7nm FinFET	7nm	5nm	7nm	14nm
Maximum operating temperature	100°C	105°C	105°C	100°C	100°C	90°C	N/A	N/A	N/A	100°C
Link	https://ark.intel.com/content/www/us/en/ark/products/213800/intel-core-i9-11980hk-processor-24m-cache-up-to-5-00-ghz.html	https://www.amd.com/en/products/apu/amd-ryzen-9-5900hx	https://www.amd.com/en/products/apu/amd-ryzen-9-5980hs	https://ark.intel.com/content/www/us/en/ark/products/213803/intel-core-i7-11800h-processor-24m-cache-up-to-4-60-ghz.html	https://ark.intel.com/content/www/us/en/ark/products/213799/intel-core-i7-11850h-processor-24m-cache-up-to-4-80-ghz.html	https://www.amd.com/en/products/cpu/amd-ryzen-7-5800x	https://www.qualcomm.com/products/snapdragon-8cx-gen-2-5g-compute-platform	https://www.apple.com/newsroom/2020/11/apple-unleashes-m1/	https://www.mediatek.com/products/tablets/mit8192	https://www.intel.in/content/www/in/en/products/sku/201909/intel-xeon-w10885m-processor-16m-cache-up-to-5-30-ghz/specifications.html

software. Moreover, the engine aims for a human-like understanding and behaviour to IoT devices.

More advanced technologies

Along with the above-mentioned functionalities, most modern processors include advanced proprietary technologies that make them better in certain aspects of work. Following are some advanced innovative technologies that are used in many latest processors.

Enhanced image processing units.

The complementary metal-oxide semiconductor (CMOS) image sensors of the mobile camera do not sense

red, green, and blue wavelengths for each pixel. It only senses one colour for each. Now, it is the job of image signal processors (ISPs) to interpolate this image by applying methods of computing. In addition, the ISP does all other housekeeping; it controls autofocus, exposure, and white balance for the camera system. Moreover, modern image processors can correct lens imperfections as well as things like HDR recombining, noise reduction, other filtering, face or object detection, and conversion between colour spaces.

The latest Intel i9 processors integrate an integrated ISP with

advanced hardware implementation that improves image and video quality of cameras. The Apple M1 features Apple's latest ISP for higher-quality video with better noise reduction, greater dynamic range, and improved auto white balance.

Advanced security features. The significance of security features is that they deal with today's most sophisticated attacks and viruses. The memory protection systems together provide real-time encryption of system memory for protection against physical attacks.

Latest laptop processors include